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Blockchain-Based Secure Voting System

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Abstract – Democratic governance fundamentally depends on elections that are conducted with fairness, accuracy, and public accountability. Existing voting mechanisms, whether physical ballots or centralized digital platforms, consistently fall short of these requirements due to vulnerabilities in data integrity, centralized administrative control, and the complete absence of independent voter verification. This paper presents a decentralized electronic voting platform constructed on the Ethereum blockchain, engineered to resolve these persistent failures through distributed ledger technology and advanced cryptographic design. Each ballot submitted through the system is sealed as a tamper resistant blockchain transaction, governed and validated by Solidity smart contracts that operate autonomously without any supervising authority. Voter eligibility is confirmed using Zero Knowledge Proof protocols, allowing each participant to demonstrate their right to vote without revealing personal information at any stage of the process. The complete election pipeline, covering registration, authentication, ballot submission, vote recording, autonomous tallying, and public result verification, is managed entirely within the decentralized architecture. The application layer is built using React.js for the voter interface, Node.js and Python Flask for backend processing, and IPFS for distributed off chain data storage. Security testing confirmed full resistance to duplicate voting and identity fabrication attacks. Comparative evaluation demonstrates measurable superiority over all conventional voting models across security, transparency, and verifiability. The system offers a practical foundation for trustworthy elections at organizational and national scale.

Keywords: Ethereum Blockchain, Smart Contracts, Solidity, Zero Knowledge Proofs, Decentralized Voting, Voter Anonymity, IPFS, Election Security, Distributed Ledger, Cyber Security.

I. Introduction

The right to vote is the most fundamental instrument of democratic participation, yet the infrastructure that supports this process has remained vulnerable to manipulation, error, and systemic distrust for generations. Paper ballot systems are subject to physical tampering, counting errors, and logistical failures that undermine confidence in results. Electronic voting machines, introduced to modernize the process, have introduced new categories of risk including firmware vulnerabilities, unauthorized remote access, and the concentration of critical data in systems controlled by a single administrative body. In both cases, voters are left with no mechanism to independently confirm that their vote was accurately recorded and counted. The emergence of distributed ledger technology presents a structurally different approach to this problem. A blockchain distributes transaction records across a network of independent nodes, each of which validates and stores a complete copy of the ledger. No single participant controls the network, and no record can be altered once confirmed without invalidating the entire chain of subsequent entries. These properties directly address the centralization and tamper resistance problems that undermine conventional voting systems. This paper constructs a fully operational voting platform on the Ethereum blockchain. Smart contracts written in Solidity govern every stage of the election process

automatically. Zero Knowledge Proofs preserve voter privacy while confirming eligibility. IPFS handles distributed storage of registration data. A React.js frontend makes the system accessible to any voter with an internet connection. The contributions of this work are as follows:

A fully decentralized election architecture using Ethereum smart contracts that eliminates dependence on any central authority throughout the entire voting process.

A Zero Knowledge Proof based voter authentication system that verifies eligibility without exposing voter identity to any participant on the network.

An IPFS integrated storage design that keeps sensitive registration data off chain while maintaining cryptographic proof of its integrity.

A complete full stack implementation including React.js, Node.js, and Python Flask that delivers a functional voter interface deployable across web and mobile platforms.

Section II surveys relevant prior research. Section III details the system architecture and methodology. Section IV presents the technology stack. Section V reports performance and security evaluation findings. Section VI concludes with future development directions.

II. Literature Survey

Research into technology driven improvements for electoral systems has expanded substantially over the past decade. Blockchain has emerged as the most actively investigated approach for resolving the core weaknesses of both paper and electronic voting.

Ayed [1] produced one of the earliest academic investigations into applying Ethereum smart contracts to the problem of vote management and result computation. The study established a theoretical framework demonstrating that contract based automation could replace centralized counting processes, but the absence of a working prototype and the lack of any adversarial testing left practical questions about robustness and scalability unanswered.

Agora [2] advanced beyond theoretical work by deploying a blockchain assisted verification layer alongside the 2018 Sierra Leone presidential election. The deployment offered early evidence that distributed ledger technology could function within a real electoral environment, though its role was limited to post election verification rather than primary vote recording. Concerns were also raised regarding accessibility in areas with limited connectivity infrastructure.

McCorry et al. [3] designed and tested the Open Vote Network, which implemented a fully self tallying election protocol on Ethereum that required no external authority to access individual vote data at any point. Privacy was guaranteed through the protocol structure rather than administrative policy, which represented a significant conceptual advance. The limitation of the approach was the substantial computational load it imposed on the network, restricting practical use to elections with small participant counts.

Tarasov and Tewari [4] examined the application of zk SNARK cryptography to the specific challenge of verifying voter eligibility anonymously on a blockchain. Their analysis confirmed the mathematical soundness of the approach while identifying proof generation time as a performance constraint requiring further optimization before large scale deployment would be practical.

Hjalmarsson et al. [5] evaluated Hyperledger Fabric, a permissioned blockchain framework, as an alternative infrastructure for election systems. The implementation achieved lower latency and reduced transaction costs compared to public Ethereum deployments, but these gains came at the cost of restricting network participation to a defined set of approved nodes, reintroducing a form of centralized control that undermines the trust guarantees the approach was designed to provide.

Zheng et al. [6] conducted a comprehensive technical review of blockchain infrastructure, identifying transaction throughput and network scalability as the primary obstacles to widespread adoption. The finding that Layer 2 network architectures offer a viable path to addressing both constraints was

directly incorporated into the deployment strategy of the system proposed in this paper.

The collective findings of existing research identify a clear gap in the available literature. No prior system combines cryptographic voter privacy, on chain vote verification, and a complete publicly accessible application interface within a single deployable platform. This work is specifically designed to close that gap.

III. Proposed Methodology

The proposed system delivers a complete decentralized election platform in which no individual or organization, including the party administering the election, holds the authority to alter any recorded vote, manipulate the counting process, or access voter identity information. All operational logic is embedded in smart contracts that execute identically for every participant and cannot be modified after deployment.

A. Operational Workflow

Stage 1: Voter Registration and Identity Verification. Before the election opens, eligible participants submit their credentials to the backend system. The ZoKrates framework generates a unique Zero Knowledge Proof for each approved voter that encodes their eligibility status without including any identifying information. This proof and its associated parameters are stored on IPFS under a content addressed identifier, and only the commitment hash of the proof is recorded on chain during the registration process.

Stage 2: Encrypted Ballot Submission. During the active voting period, a registered voter opens the React.js application, authenticates using their registered credentials, and selects their preferred candidate. The selection is encrypted using Elliptic Curve Cryptography before transmission. The encrypted ballot and the voter's Zero Knowledge Proof are submitted together as a single signed transaction to the Ethereum smart contract, which validates the proof and checks the voter's submission status before accepting the ballot.

Stage 3: Immutable Vote Recording. Every validated ballot transaction is written permanently to the Ethereum blockchain. Each recorded entry carries a cryptographic hash linking it to the preceding block in the chain. The chained hash structure ensures that altering any historical record would invalidate every subsequent block, making retroactive modification computationally infeasible on any live network.

Stage 4: Autonomous Result Computation. At the programmed election closing time, the smart contract independently tallies all valid recorded ballots and computes the result for each candidate. No human action is required to trigger or oversee this process. The computation is executed entirely by the contract code and the resulting tally is immediately published as a permanent on chain record.

Stage 5: Open Public Verification. Following result publication, the complete transaction history of the election remains permanently accessible on the public blockchain. Any party can independently verify the total vote count, confirm that every recorded transaction followed the correct protocol, and validate that the published result accurately reflects all recorded data. No transaction can be connected back to the individual voter who submitted it.

Fig. 1: Proposed System Architecture and Workflow

B. System Architecture

Voter Interaction Layer. A React.js single page application provides the primary interface for voters and election administrators. Web3.js libraries handle all communication between the browser application and the deployed Ethereum contracts, enabling direct blockchain interaction without any intermediary service.

Backend Processing Layer. Node.js with Express manages API routing and voter registration workflows. Python with Flask handles Zero Knowledge Proof generation through the ZoKrates engine and coordinates all interactions with IPFS for off chain data storage.

Smart Contract Execution Layer. Solidity contracts deployed on the Ethereum virtual machine encode and enforce all election rules. Contracts govern voter registration, ballot acceptance, duplicate prevention, result computation, and public result access. Once deployed, no party can alter the contract code or the rules it enforces.

Blockchain Transaction Ledger. The Ethereum blockchain serves as the permanent, authoritative record of every vote cast. The chained hash structure of the ledger means each confirmed block implicitly validates the complete preceding transaction history.

Distributed File Storage. IPFS stores voter eligibility records, registration proofs, and ZKP parameters off chain. Content addressing on IPFS ensures that any modification to stored data produces a different content identifier, making silent alteration impossible.

C. Zero Knowledge Proof Integration

The voter privacy guarantee rests on zk SNARK proofs generated through ZoKrates and verified on chain by a Solidity verifier contract. During registration each voter receives a unique proof encoding their eligibility. When submitting a ballot the voter includes this proof in the transaction. The on chain verifier checks the mathematical validity of the proof against the registered commitment hash. A valid, previously unused proof results in the ballot being accepted. Any fabricated or reused proof is rejected. This mechanism guarantees that every accepted vote comes from a genuinely eligible voter who has not previously voted, while ensuring that voter identities remain inaccessible to any network participant.

D. Smart Contract Function Design

The core election contract exposes four primary functions:

- **registerVoter(address voter, bytes32 zkProofHash):** Records the ZKP commitment hash for each eligible voter during the pre election setup phase. Locked against new entries once the election begins.
- **castVote(uint candidateId, bytes memory proof):** Accepts an encrypted ballot alongside its associated proof. Verifies the proof, confirms the voter has not previously submitted a ballot, and atomically records the vote if all conditions are satisfied.
- **tallyVotes():** Executable only after the election closing timestamp encoded in the contract at deployment. Computes and publishes the final candidate totals as a permanent on chain event.
- **getResults():** A read only query function returning current vote totals to any caller at any time after tallying, requiring no transaction and incurring no gas cost.

IV. Technology Stack: Tools and Platforms

The platform was constructed using technologies selected to satisfy the security, scalability, and accessibility requirements of a production grade election system. Table I summarizes each component and its function within the overall architecture.

Table I: Technology Stack Summary

Component	Technology	Role in System
Blockchain Network	Ethereum and Polygon	Permanent decentralized vote transaction storage
Contract Language	Solidity v0.8+	Election rule encoding and autonomous enforcement
Development Environment	Truffle and Hardhat	Contract compilation, testing, and deployment

Voter Interface	React.js with Web3.js	Browser based voting application and chain communication
Backend Services	Node.js Express and Python Flask	API routing, proof generation, and storage management
Off Chain Storage	IPFS	Distributed storage of registration data and ZKP parameters
Privacy Layer	ZoKrates and snarkjs	zk SNARK circuit construction and on chain verification
Vote Encryption	Elliptic Curve Cryptography and SHA 256	Ballot confidentiality and data integrity verification
Contract Testing	Mocha and Chai	Automated functional and integration test execution
Network Connectivity	Infura and Alchemy	Remote Ethereum node access for deployment and monitoring

Ethereum was selected as the primary network for its mature smart contract tooling, established security record, and compatibility with Layer 2 scaling solutions. Polygon was designated as the large scale deployment target because it processes transactions in under two seconds at significantly lower cost while accepting the identical Solidity contract code without modification.

ZoKrates was chosen specifically because it generates deployable Solidity verifier contracts directly from circuit definitions, keeping the entire proof verification process on chain with no dependency on external services. IPFS was incorporated to manage the cost and scalability constraints of on chain data storage while preserving cryptographic integrity guarantees through content addressing.

V. Result Analysis and Performance

The system was evaluated through simulated elections on the Ethereum Sepolia testnet with voter populations of 100, 500, and 1000 participants. All smart contract logic was verified using an automated suite built with Truffle, Mocha, and Chai.

A. Security Evaluation

Double submission tests confirmed that the contract rejected 100% of attempts to cast a second ballot from any address that had already submitted a vote, enforced through the hasVoted mapping checked before each transaction. Identity fabrication tests showed that every attempt to construct and submit a fraudulent Zero Knowledge Proof was rejected by the on chain verifier, as the mathematical relationship between proof and registered commitment hash cannot be reproduced without the original private inputs. Post election data integrity checks confirmed that no previously recorded transaction had been altered in any form, validating the tamper resistance of the blockchain ledger under all tested conditions.

B. Comparative Evaluation

Table II compares the proposed system against three conventional voting approaches across five criteria considered essential for a trustworthy election.

Table II: Comparative Evaluation of Voting Systems

Voting Approach	Security	Transparency	Voter Privacy	Tamper Resistance	Verifiability
Paper Based Voting	Low	Low	Partial	No	No
Electronic Voting Machine	Medium	Low	High	Partial	No
Centralized Online Voting	Medium	Low	High	No	No
Proposed Blockchain System	Very High	Very High	Complete via ZKP	Yes	Full End to End

C. Performance Measurements

Vote transactions on the Ethereum Sepolia testnet confirmed in an average of 13 seconds. The identical transactions on the Polygon Mumbai testnet confirmed in an average of 1.7 seconds. Client side ZKP generation averaged 1.8 seconds on standard consumer hardware. On chain proof verification consumed approximately 450,000 gas units per transaction. Initial contract deployment required approximately 3.2 million gas units. These figures confirm that Ethereum mainnet deployment is viable for elections up to 10,000 voters and that Polygon is the appropriate choice for larger scale deployments.

D. Testing and User Evaluation

All 42 automated contract test cases passed without exception, covering every function and every defined failure condition. End to end integration tests confirmed correct operation across the full path from browser input to confirmed blockchain transaction. A usability study with 15 student and faculty participants produced an average satisfaction rating of 4.3 out of 5. Participants consistently rated the platform higher than conventional voting tools on trustworthiness and result transparency while describing the interface as accessible without any prior blockchain knowledge.

VI. Conclusion

This paper presented a fully functional decentralized voting platform built on the Ethereum blockchain, designed to eliminate the three fundamental weaknesses of existing election infrastructure: susceptibility to data manipulation, dependence on centralized trust, and the absence of independent vote verification. By encoding all election rules in immutable Solidity smart contracts, recording all ballots as permanent blockchain transactions, and using Zero Knowledge Proofs to protect voter identity throughout the process, the system delivers security, transparency, and privacy simultaneously, a combination no existing voting model currently provides.

Security evaluation confirmed complete prevention of duplicate voting and identity fabrication. Performance testing demonstrated practical viability for real elections, particularly on Layer 2 networks. User evaluation confirmed platform accessibility for participants without technical backgrounds.

Future development will pursue biometric voter verification using fingerprint and facial recognition, optimization of ZKP circuit design to reduce on chain gas consumption, deployment at national scale on Polygon zkEVM, integration of IoT connected physical polling stations for hybrid election environments, multilingual interface support for diverse voter populations, and compatibility with government issued digital identity frameworks such as Aadhaar for use in official national elections.

REFERENCES

[1] A. B. Ayed, "A Conceptual Secure Blockchain Based Electronic Voting System," International Journal of Network Security and Its Applications, vol. 9, no. 3, pp. 1 9, May 2017.

[2] Agora, "Agora: Blockchain Voting in Sierra Leone," Technical White Paper, 2018. [Online]. Available: https://www.agora.vote

[3] P. McCorry, S. F. Shahandashti and F. Hao, "A Smart Contract for Boardroom Voting with

Maximum Voter Privacy," in Proc. 21st International Conference on Financial Cryptography and Data Security, Malta, 2017, pp. 357 375.

[4] P. Tarasov and H. Tewari, "The Future of E Voting," IADIS International Journal on Computer Science and Information Systems, vol. 12, no. 2, pp. 148 165, 2017.

[5] F. J. Hjalmarsson, G. K. Hreidarsson, M. Hamdaqa and G. Hjalmrtysson, "Blockchain Based E Voting System," in Proc. IEEE 11th International Conference on Cloud Computing, San Francisco, CA, USA, 2018, pp. 983 986.

[6] Z. Zheng, S. Xie, H. N. Dai, X. Chen and H. Wang, "Blockchain Challenges and Opportunities: A Survey," International Journal of Web and Grid Services, vol. 14, no. 4, pp. 352 375, 2018.

[7] V. Buterin, "Ethereum: A Next Generation Smart Contract and Decentralized Application Platform," Ethereum Foundation White Paper, 2014. [Online]. Available: https://ethereum.org/en/whitepaper/

[8] E. Ben Sasson, A. Chiesa, C. Garman, M. Green, I. Miers, E. Tromer and M. Virza, "Zerocash: Decentralized Anonymous Payments from Bitcoin," in Proc. IEEE Symposium on Security and Privacy, San Jose, CA, USA, 2014, pp. 459 474.

[9] J. Groth, "On the Size of Pairing Based Non Interactive Arguments," in Proc. Annual International Conference on the Theory and Applications of Cryptographic Techniques, Vienna, Austria, 2016, pp. 305 326.

[10] S. Nakamoto, "Bitcoin: A Peer to Peer Electronic Cash System," Self Published White Paper, 2008. [Online]. Available: https://bitcoin.org/bitcoin.pdf

[11] J. Benet, "IPFS: Content Addressed, Versioned, Peer to Peer File System," arXiv preprint arXiv:1407.3561, 2014.

[12] N. Kshetri and J. Voas, "Blockchain Enabled E Voting," IEEE Software, vol. 35, no. 4, pp. 95 99, 2018.

[13] I. Bashir, Mastering Blockchain: Distributed Ledger Technology, Decentralization and Smart Contracts Explained, 2nd ed., Packt Publishing, Birmingham, UK, 2018.

[14] R. Hardwick, A. Gioulis, R. Akram and K. Markantonakis, "E Voting with Blockchain: An E Voting Protocol with Decentralisation and Voter Privacy," in Proc. IEEE International Conference on Internet of Things, Halifax, Canada, 2018, pp. 1561 1567.

[15] S. Wang, L. Ouyang, Y. Yuan, X. Ni, X. Han and F. Y. Wang, "Blockchain Enabled Smart Contracts: Architecture, Applications and Future Trends," IEEE Transactions on Systems, Man and Cybernetics: Systems, vol. 49, no. 11, pp. 2266 2277, Nov. 2019.